

SIMULATION PROTOCOL: BEGINNER OVERVIEW



TOP HANKOOK POWDER MACHINE ENGINEERING

Fine Powder Technology & Sustainable Recycling Systems

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Fine Powder Technology & Sustainable Recycling Systems



STATUS: ONLINE // LOADING ASSETS...

SYSTEM HUB // IDENTITY

Fine Powder Technology Experts specializing in fully automated industrial powder systems.

SYSTEM HUB // SCALE

Comprehensive engineering from precision Laboratory Scale testing to Heavy Industrial manufacturing lines.

SYSTEM HUB // MISSION

Values Resource & Environment — Leading the global transition to sustainable EV & LI-ion Battery Recycling.



MODULE 01 // PARTICLE REDUCTION

Accessing precision milling and
micro-grinding technologies...



 [SYSTEM READDUT:
KINETIC MODELS LOADED]

ASSET: TURBO MILL

Capacity: 20–1000 kg/hr | Target Particle Size: 10–150µm

Horizontal Form Factor

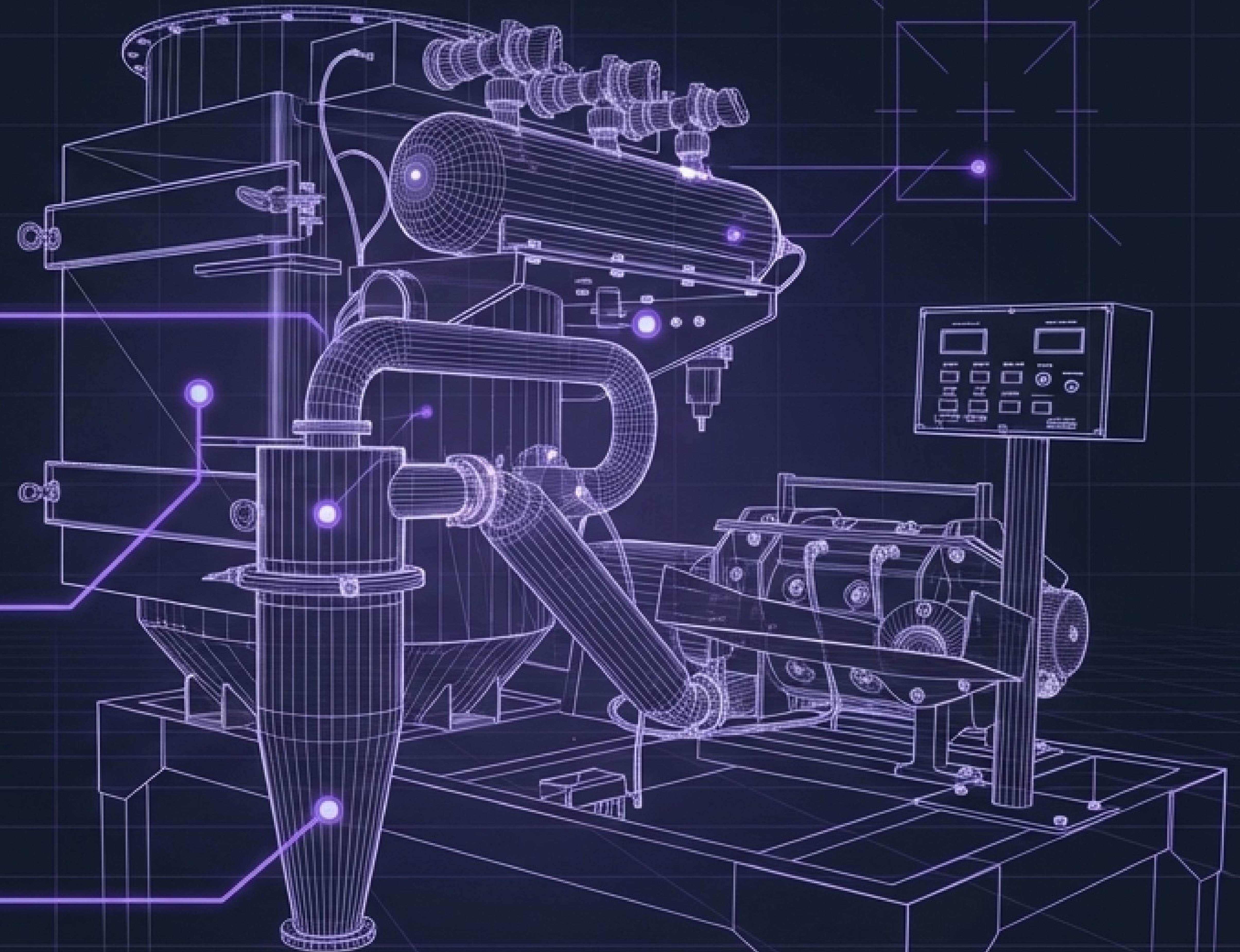
Special spiral rotor guarantees maximum rotational stability.

Screen-less Architecture

Executes grinding and classifying simultaneously without physical screens.

Thermal Control

Chamber engineered to minimize internal heat rise during high-friction operations.



TURBO MILL // KINETIC PRINCIPLE



Ø1 - INJECTION

Raw material is fed directly into the machine via the central inlet.



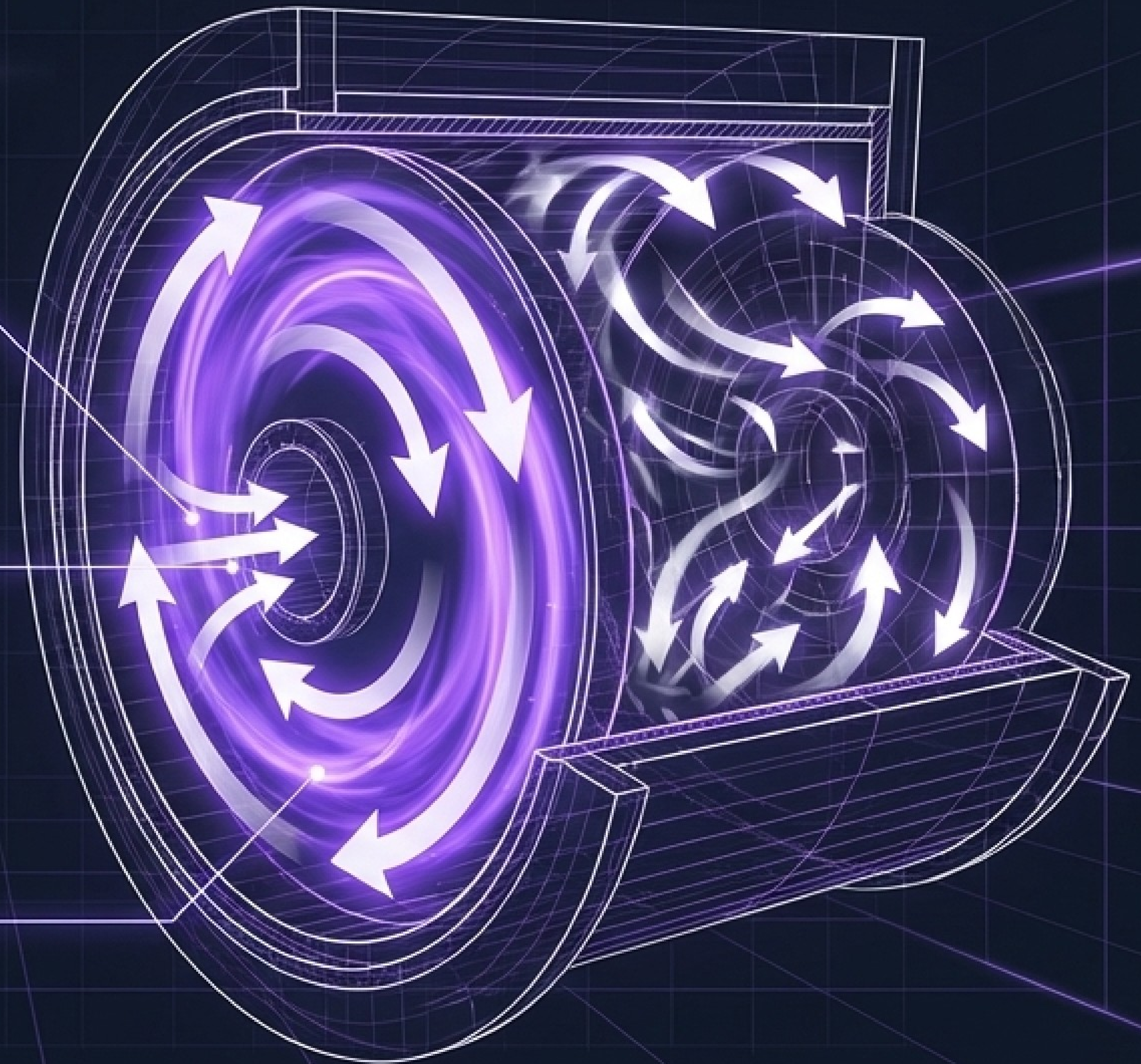
Ø2 - CENTRIFUGAL FORCE

A high-speed rotating impeller captures the material and throws it outward against a rigid grinding liner.



Ø3 - HYDRAULIC AIR FLOW

High-turbulent internal air currents pulverize the particles through intense mutual collision and shear force.



ASSET: ROLL MILL

TARGET PROFILE:

Engineered specifically for heat-sensitive materials & high-precision chemical operations.

Dual Force Mechanism

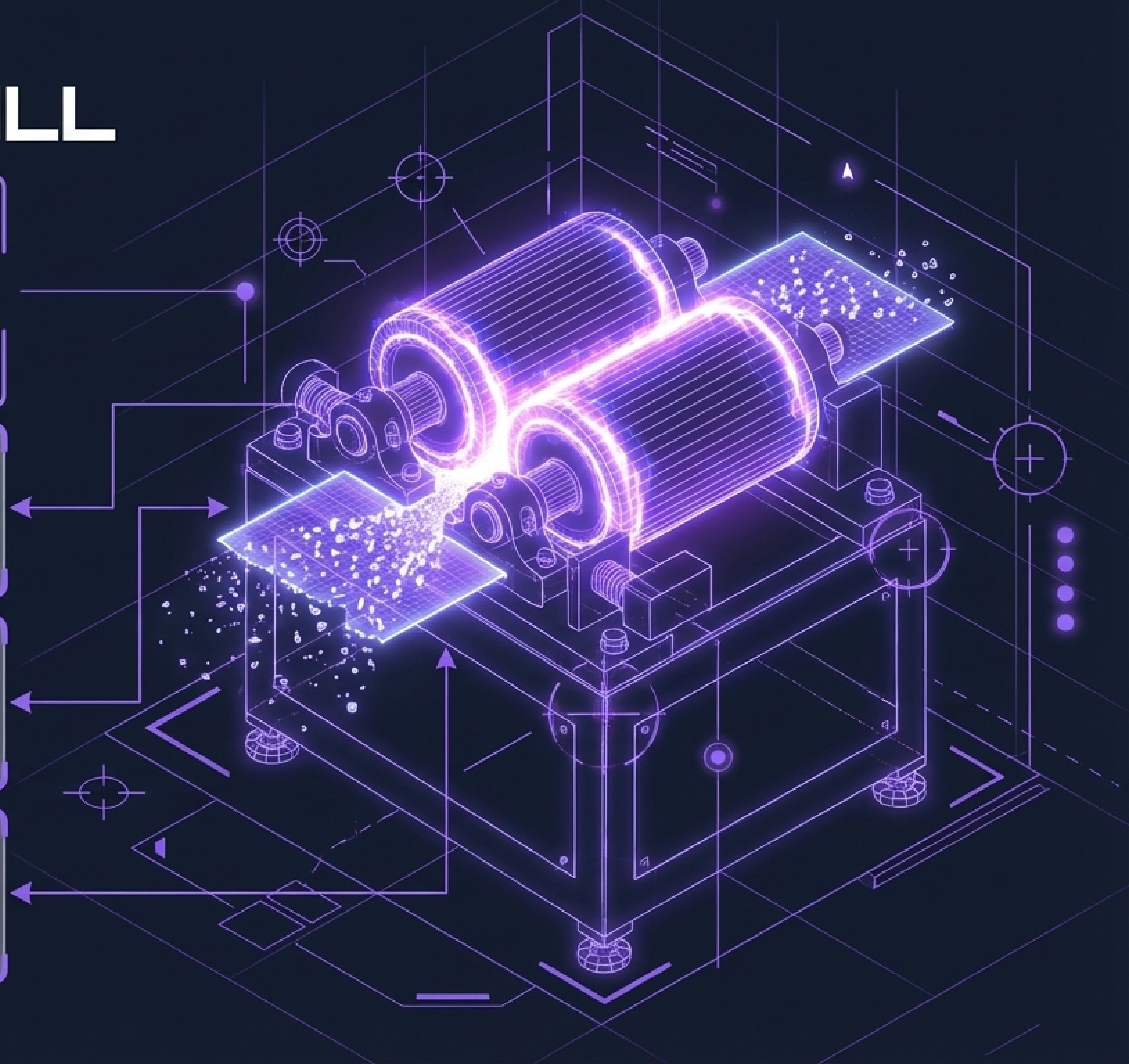
Combines direct impact with sustained friction.

Low Velocity / Low Output

Operates at an ultra-quiet 50~500m/min, minimizing noise and vibration.

Real-time Calibration

Grinding gap is continuously adjustable without halting operation.



DIAGNOSTIC MATRIX // MILLING COMPARISON



TURBO MILL

ROLL MILL

Primary
Kinetic Force

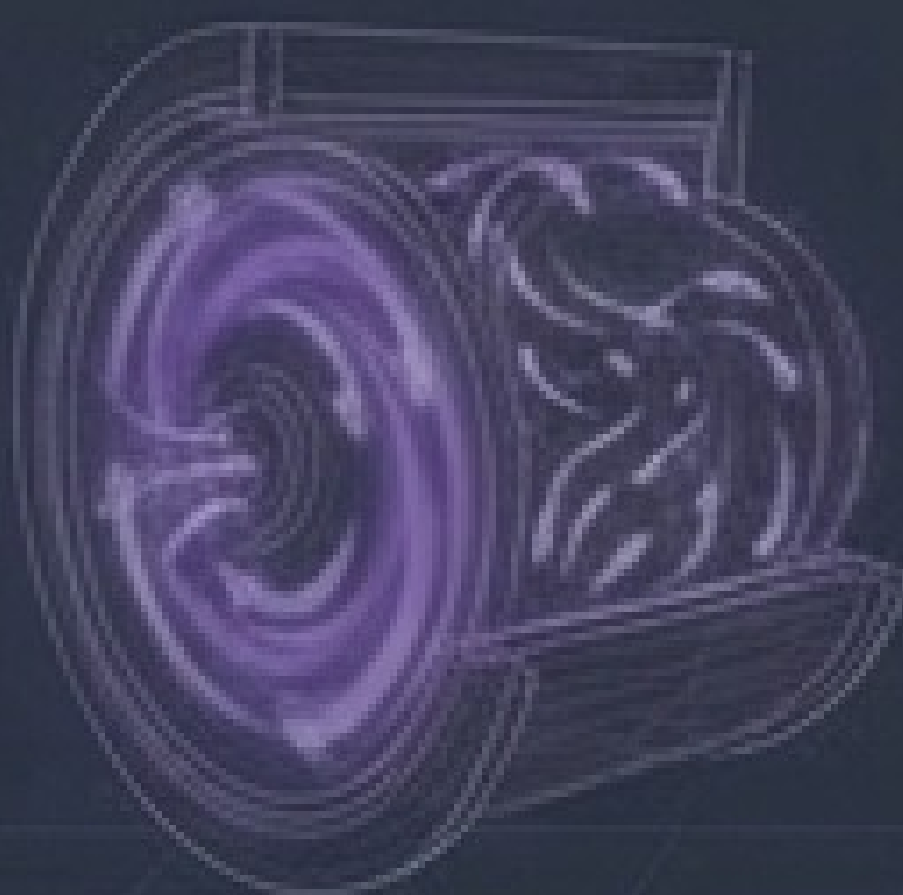
High-speed air turbulence &
Centrifugal force.

Low-speed compression & Friction.

Machine
Architecture

Single high-speed enclosed chamber.

Dual-rotating, heavy-duty cylinders.

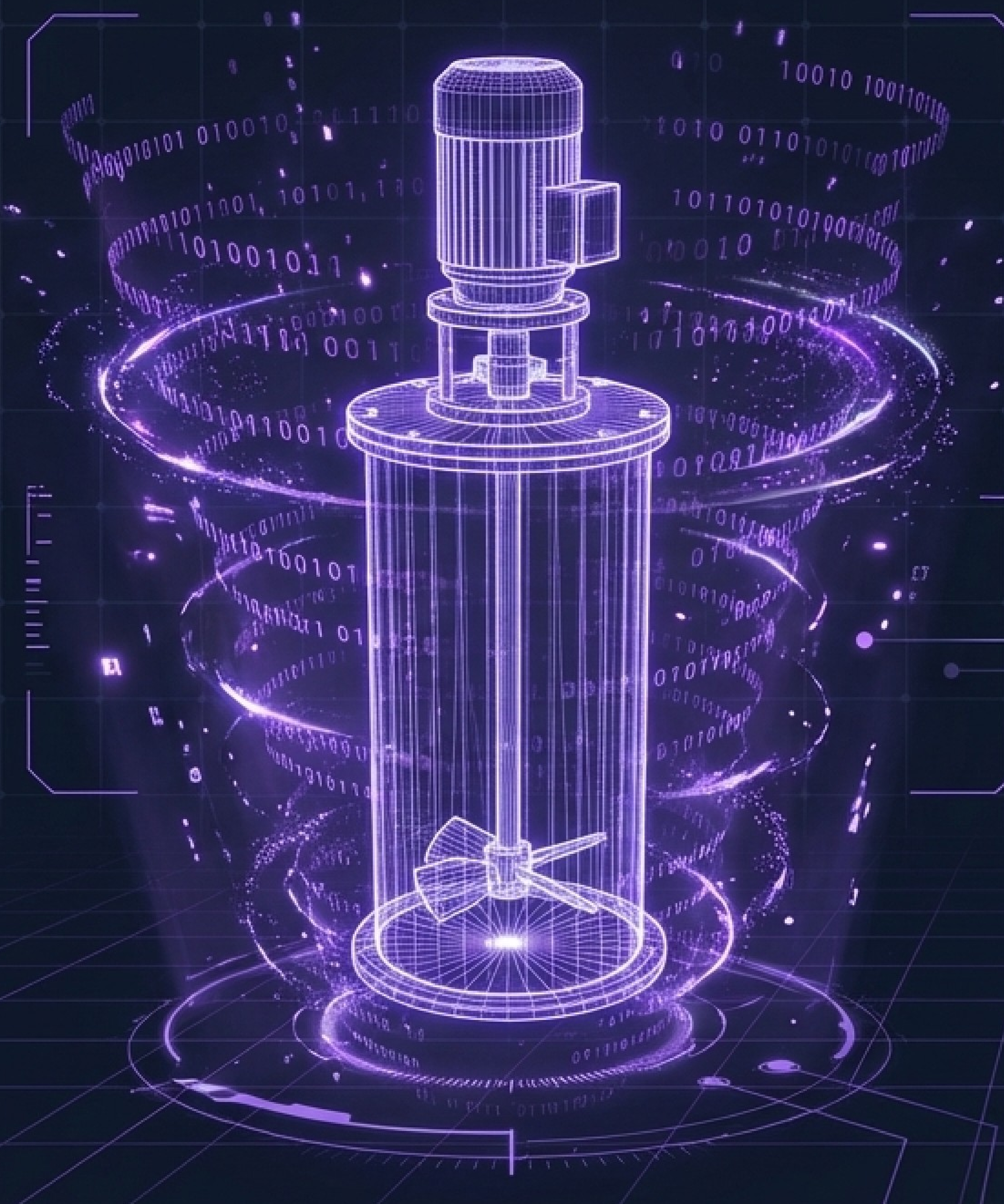


Ideal Use
Case

Ultra-fine grinding (10-150 μ m),
ceramics, non-ferrous metals.

Heat-sensitive materials, chemicals,
paints, food processing.





MODULE 02 // SYNTHESIS & BLENDING

ASSET: HIGH SPEED MIXER



[Core Function]

Executes blending, dispersing, and granulating in one rapid, unified process.



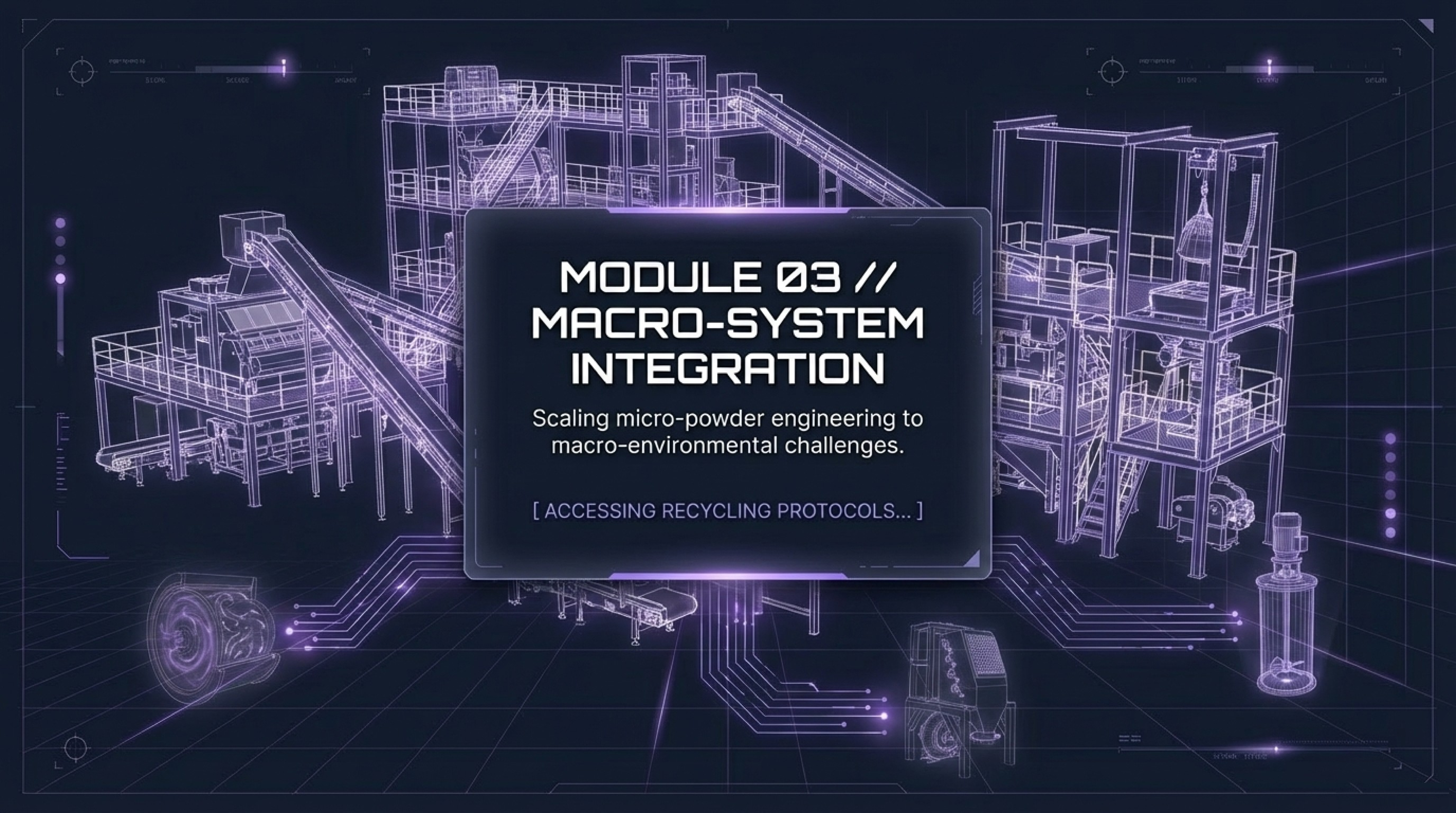
[Kinetic Mechanism]

High peripheral tip speed of the central impeller generates intense shearing and friction forces.



[Versatility Matrix]

Capable of seamless cross-processing for both WET (liquids/emulsification) and DRY (powders/granules) materials.



MODULE 03 //

MACRO-SYSTEM INTEGRATION

Scaling micro-powder engineering to
macro-environmental challenges.

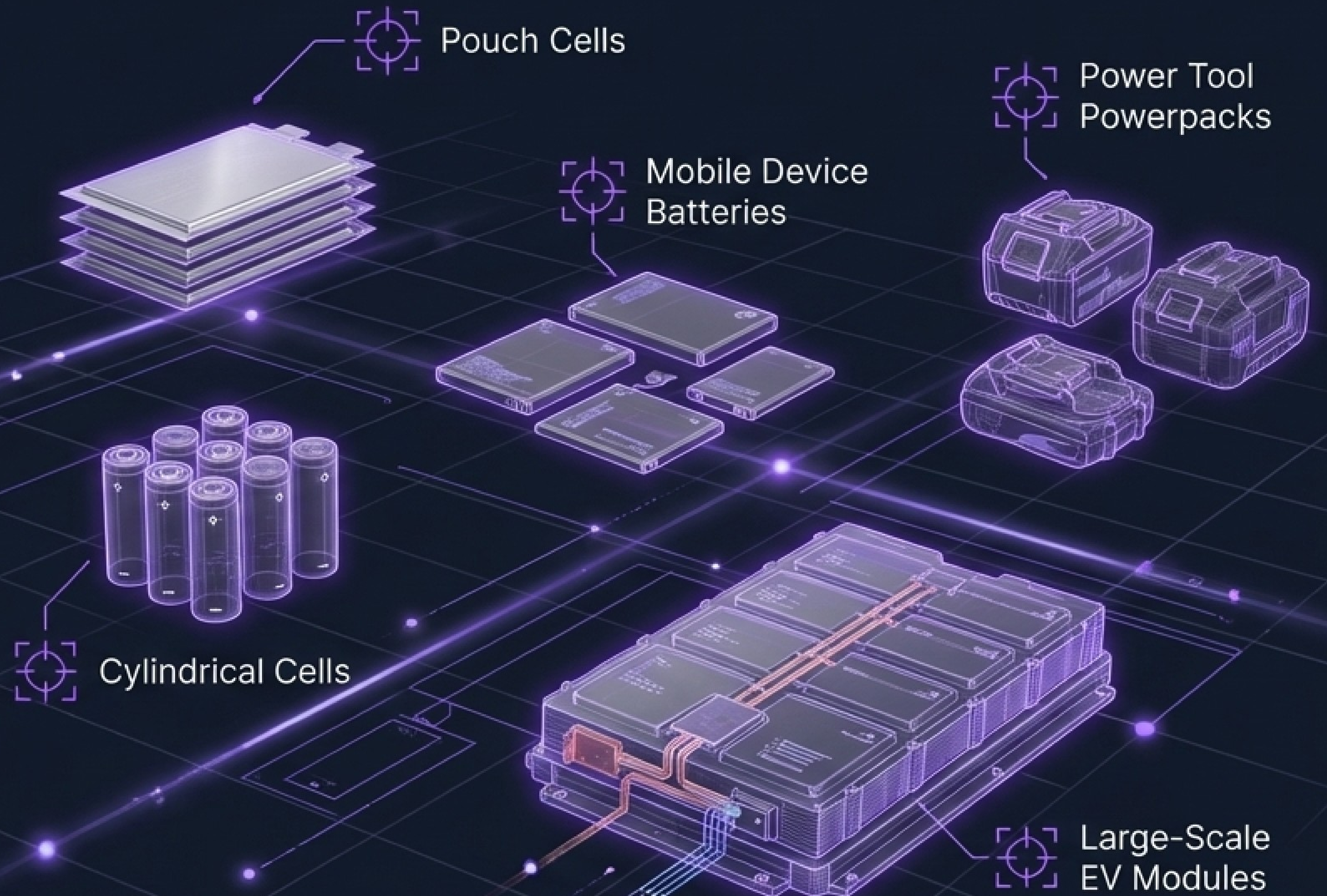
[ACCESSING RECYCLING PROTOCOLS...]

SYSTEM // LITHIUM-ION RECYCLING PROTOCOL

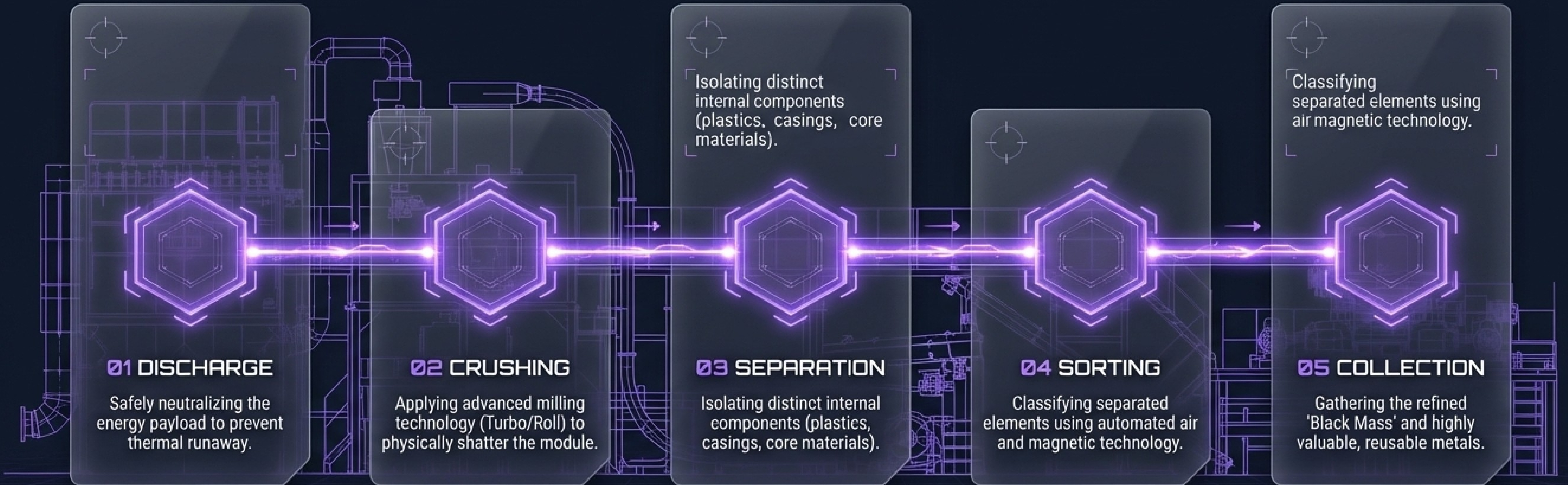
CORE DIRECTIVE: Values Resource & Environment

PROCESSING CAPABILITIES NODE:

Engineered to handle all modern spent lithium battery architectures:



THE E.V. BATTERY RECYCLING PIPELINE



System Blueprint Dimensions:
26.0m (L) x 9.1m (H) x 6.1m (W)

PROTOCOL COMPLETE // THE CLOSED LOOP



TOP Hankook's automated powder technology ensures the hazardous waste of today is meticulously transformed into the high-grade raw materials of tomorrow.

SYSTEM LOGOUT // SECURE COMM-LINK

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